

Methods to enhance the guaranteed tensile strength of GFRP rebar to 900 MPa with general fiber volume fraction



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HIGHLIGHTS

- Some factors influencing the tensile strength of GFRP rebar were examined.
- Filler type may affect the tensile strength of GFRP rebar, yet resin type does not.
- Pre-tensioning fibers during manufacturing can decrease the voids in the cross-section.
- Decreased voids and aligned fibers increase tensile strength.
- GFRP rebar shows a tensile strength of up to 900 MPa with 78% fiber volume fraction.

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ABSTRACT

The corrosion of reinforcement (rebar) in concrete members is one of the governing factors deteriorating the load carrying capacity of concrete members. Fiber reinforced polymer (FRP) has been the subject of much interest from researchers and engineers due to its enhanced durability and non-corrosive characteristics. Corrosion problems in reinforced concrete (RC) members can be resolved through the use of FRP rebar as its reinforcement. In order to expand usage of FRP rebar, rebar with various shapes and sizes has been developed and manufactured for RC structures. However, currently, a limited number of FRP rebar selections are available on the market. The obstacles preventing the wider application of FRP rebar are the relatively high price of FRP rebar compared to steel rebar and insufficient design guidelines for various FRP rebars sold in the market. Since there is a wide variety of unstandardized FRP rebar sold in the market, the application of these rebars in concrete structural design must be carefully conducted by modifying related coefficients and safety factors to assure the safety of the designed structures. Therefore, the enhancement of the performance of FRP rebar by means of efficient manufacturing methods would reduce its cost and increase its usage. Sufficient tensile and bonding strengths are necessary properties of rebar. However, depending on the material and shape of FRP rebar, these properties vary significantly. This study focuses on the tensile performance of rebar realized through enhancements in its constituent materials and manufacturing processes. Constituent materials and manufacturing processes were thoroughly researched and evaluated to maximize the tensile performance. A total of six factors affecting the tensile and bonding capacities of FRP rebar were modified in the study. For the new FRP rebar, a higher target with a guaranteed tensile strength of 900 MPa and a 78% fiber volume fraction by weight, manufactured using E-glass fiber TEX 4400, was chosen. The verification study results showed that the new glass FRP (GFRP) rebar showed better material properties than other types of GFRP rebar currently available on the market.

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1. Introduction

According to ACI committee 440 [1], corrosion problems in RC structures began in the 1960s. Corrosion byproducts cause a vol-

ume expansion around corroded rebar, resulting in the deterioration of the concrete. A galvanized coating was the first solution to the problem, but it did not become widely used for a variety of reasons. Epoxy-coated steel rebar was used in the early 1970s based on an evaluation of over 50 types of coating materials for steel rebar. Meanwhile, there were several attempts to make rebar using non-corrosive materials, as corrosion is not a structural but a

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material problem. Fiber reinforced polymer (FRP) is a composite material made of a polymer matrix reinforced with fibers. The fibers usually consist of glass, carbon, basalt or aramid. The polymer matrix is usually epoxy, vinyl-ester or a polyester thermosetting plastic. FRPs have been used for decades in the aeronautical, aerospace, automotive, and marine industries due to their inherent advantages of high strength, low weights, and non-corrosive properties. FRP products are manufactured in many different forms, such as bars, sheets, or 2D grids. Their usage in civil engineering fields dates back to the 1950s, when glass FRP (GFRP) rebar was first investigated for structural use. However, it was not until the 1970s that FRP was finally used in structural applications. FRP rebar of various shapes and sizes has been developed and is now commercially available [2]. However, its application to concrete structures is still limited, especially as a main structural component.

The major problems which arise when using FRP as a rebar material is its high price, low elastic modulus, and more brittle failure behavior than steel rebar. Even though it is possible to improve the elastic modulus of FRP bar by hybridization with fiber with high elastic modulus, this would lead to a price increase of FRP rebar, as high-modulus fibers are generally expensive [3]. Many solutions are available for improving the performance of FRP rebar (e.g., simply using fibers with higher strength or a larger amount of fibers). However, in order to be competitive in the reinforcement market as structural rebar, the most important feature of FRP rebar is its price-to-performance ratio. A reduction in the price is possible either by reducing the production cost while maintaining the performance or improving the performance while maintaining the cost. Moreover, unlike steel reinforcement, the tensile properties of the FRP rebar depend on the content and size of the member. In particular, when a tensile force is applied to a larger diameter rebar, the tensile strength experiences a size effect wherein higher stresses develop in the outer fibers than in the inner ones [4]. Unfortunately, the size effect of the rebar was not considered in this study. However, due to the importance of the topic, effect of rebar size will be studied in-depth in the future work.

In this study, the approach of enhancing the performance of GFRP rebar manufactured using the materials and conditions currently used in the industry while maintaining the production cost is taken. The tensile strength of GFRP rebar is defined as a guaranteed tensile strength equal to the result of subtracting three times the standard deviation of its tensile strength from the average tensile strength. In the new GFRP, rebar with an inner diameter of 12.7 mm and a volume fraction of 78% by weight was manufactured using E-glass fiber with TEX 4400 and a vinyl-ester resin. The rib height of the rebar was excluded from the diameter, because it does not contribute to the tensile strength of the rebar. Although the current guaranteed tensile strength is less than 900 MPa, a target strength value of 900 MPa was selected here to compete with the strength of steel rebar so as to meet the structural tensile and allowable fatigue strength levels stipulated in various design codes and guidelines. For a precise understanding of the factors affecting the performance of the materials, various influential factors were initially investigated. Then, efforts to optimize the constituent materials and manufacturing processes were undertaken. Finally, tensile and bond strength tests were performed to verify the performance enhancement of the new GFRP rebar.

2. Research background

FRP rebar or composite with various shapes and properties was developed by researchers in the past, and several manufacturers are currently selling several types of rebar commercially for construction work [5,6]. For FRP composites having the shape of rebar,

Marshall Industries Composites, Inc., based in the US, produces C-Bar, which has a shape similar to that of ordinary steel rebar [7]. Hughes Brothers Inc. and Pultrall Inc., based in the US and Canada, respectively, produce Aslan 100 and V-Rod, respectively, which both use E-glass as the main constituent material, and which have helically wrapped and sand-coated surfaces to enhance the bonding. Schöck Bauteile GmbH based in Germany produces ComBAR, which has continuous spiral pitches on the surface, giving it bonding properties similar to those of deformed steel rebar [5]. Commercially available rebar with various shapes and mechanical properties are shown in Fig. 1 and Table 1 [5,8–10], respectively.

To assist with the systematic use of FRP rebar developed from various studies, the Japan Society of Civil Engineers (JSCE) and EUROCRETE published the concrete structural design guideline for FRP rebar in 1996. Then, in 1998, the Commission on Security and Cooperation in Europe (CSCE) published design recommendations for FRP RC bridges. In 2001, the American Concrete Institute (ACI) and the Canadian Network of Centers of Excellence on Intelligent Sensing for Innovative Structures (ISIS Canada) published the first version of their design recommendations for internal FRP rebar as well as a series of manuals on the use of internal/external and pre-stressed FRP rebar. In 2006, the National Research Council (CNR) published the Italian design recommendations for internal FRP rebar [11]. Presently, there are a variety of national structural design codes available for designs of FRP RC structures, but these codes still do not sufficiently consider fatigue and long-term performance of FRP RC structures [2]. FRP RC design guidelines essentially follow previous design guidelines pertaining to ordinary steel RC structures, though the coefficients and parameters used in the design equations are modified based on research results. For example, when the compression reinforcement ratio is equal to zero, ACI 318–05 [12] calculates the long-term deflection due to creep and drying shrinkage with Eq. (1).

$$\Delta_{(cp+sh)} = \xi(\Delta_i)_{sus} \quad (1)$$

Here, $\Delta_{(cp+sh)}$, $(\Delta_i)_{sus}$, and ξ represent the additional deflection due to creep and shrinkage under sustained loads, the immediate deflection due to sustained loads, and a time-dependent factor for a sustained load, respectively. Based on Eq. (1), ACI 440.1R-06 [4] gives the long-term deflection of FRP RC members by multiplying 0.6 by the ξ coefficient based on experimental results. This expression is given in Eq. (2).

$$\Delta_{(cp+sh)} = 0.6\xi(\Delta_i)_{sus} \quad (2)$$

The reason for implementing the coefficient modification approach for FRP RC design purposes stems from the lack of a consensus among structural design engineers. Due to the uncertainties



Fig. 1. Pultrall V-Rod, Hughes Bros. Aslan 100 FRP, and Schöck ComBAR (Left to Right) [5].

Table 1

Typical mechanical properties of GFRP rebars with approximately 12.7 mm diameter [8–10].

Bar name	V-ROD STD	Aslan100	C-Bar
Ultimate tensile strength (MPa)	–	–	770
Guaranteed tensile strength (MPa)	1140	758	–
Modulus of elastic (GPa)	52.5	46	42

in using FRP rebar to design concrete structures, engineers typically design FRP RC structures conservatively and implement additional safety coefficients. This approach seems logical due to the insufficient data on FRP RC structures when compared to ordinary RC structures. The ACI committee [4] recommends that the tensile strength of FRP rebar used in designs should be based on the type and level of environmental exposure, as long-term exposure to various types of environments can reduce the tensile strength, creep rupture and fatigue endurance capacity of FRP rebar (e.g., 0.7 for GFRP when concrete is exposed to earth and weather). Also, to avoid creep rupture of the FRP rebar under sustained stresses or cyclic and fatigue loadings, the ACI committee [4] recommends limiting the stress levels of FRP rebar under these stress conditions. The allowable creep rupture stress limits are 0.2 with regard to the design tensile strength for GFRP and 0.55 for carbon fiber reinforced polymer (CFRP). Therefore, a tensile strength of approximately 14–16% of the guaranteed tensile strength is used in designs [13]. When the guaranteed tensile strength of GFRP rebar is approximately 700 MPa, the design value considering the long-term behavior becomes approximately 100 MPa. In ordinary RC design codes, the allowable fatigue strength of steel rebar is defined as 125 MPa [14]. Therefore, if the environment conditions and fatigue loading induced creep rupture effect are considered, the minimal guaranteed tensile strength of FRP rebar must be 900 MPa. Even though a commercial FRP rebar produced with a diameter of 12.7 mm and a guaranteed tensile strength over 900 MPa is now available, it has a glass content of 80% by weight and is made using S-glass [5], which is more than that of the GFRP rebar having about 78% discussed in this study. S-glass has a higher tensile strength than E-glass [2]. Using higher quality glass fiber with increased fiber content also may increase the price of the rebar. It is possible to manufacture high-strength FRP rebar using high-quality constituent materials, but the production cost of these FRP rebar specimens would be too high for them to be used in construction works. Therefore, in order to promote the usage of FRP rebar, price reductions and performance upgrades through the optimization of the manufacturing process and constituent materials are the only viable options to achieve the target strength.

3. Enhancement of the efficiency of influential factors

3.1. Optimization of material composition

3.1.1. Volume fraction of fiber

The tensile properties of rebar are the most important design properties of RC and pre-stressed concrete (PSC) structures. The tensile strength (σ_{FRB}) and elastic modulus (E_{FRB}) of a longitudinally reinforced fiber composite material are estimated by including the fiber and matrix effects, as shown in Eq. (3) [15],

$$\sigma_{FRP} = \sigma_f V_f + \sigma_m V_m, \quad (3a)$$

$$E_{FRP} = E_f V_f + E_m V_m, \quad (3b)$$

where σ_f , V_f , E_f and σ_m , V_m , E_m are the corresponding tensile strengths, volume fractions, and elastic moduli of the fiber and the matrix, respectively.

If there is no void in the FRP, the sum of the volume fractions of the fiber and matrix equals 1.0. When the tensile strength of the FRP is estimated using Eq. (3), the tensile strength of the matrix is often ignored because it is very small compared to that of the fiber. Consequently, when material types for FRP are selected, a main factor governing the tensile properties of FRP is the volume fraction of the fiber. Therefore, the tensile strength of FRP rebar can be enhanced by increasing the volume fraction of the fiber without changing the cross-section. In order to have more fibers in the cross-section, it is important to remove the spaces between the fibers. Therefore, the possibility of compacting the core fibers by tightly surface-wrapping them is one feasible solution to improve the strength. This is an enhancement of the manufacturing process, which will be explained in more detail later in the paper.

3.1.2. Resin

The selection of the proper matrix is a critical requirement with regard to the manufacturing of FRP rebar, as the physical and thermal properties of the matrix significantly affect the final mechanical properties and manufacturing process of the FRP rebar [2]. The matrix not only surrounds the fibers and protects them from mechanical abrasion, but it also transfers stresses between the fibers. The matrix must be able to withstand higher ultimate strain levels than the fibers to be able to utilize the full strength of the fibers. Thermosetting and thermoplastic polymeric matrices are widely used for FRP composites. The former type is used more often, and some commonly used thermosetting polymers are polyesters, vinyl-esters, and epoxies. Changes in the tensile strength of FRP rebar with respect to the type of thermosetting polymer were investigated experimentally in this study. Glass or carbon fibers applied with unsaturated polymer or vinyl-ester resins were used to produce FRP rebar. A total of 16 specimens each 12.7 mm in diameter were manufactured and tested. The tensile strengths and elastic moduli of the specimens with various combinations of fibers and resins are shown in Table 2.

The average tensile strengths of the GFRP rebar made with vinyl-ester and unsaturated polyester was 972 and 983 MPa, respectively. When the specimens were made with carbon fiber, the average tensile strengths were 1454 and 1431 MPa for vinyl-ester and unsaturated polyester, respectively. The increased ratios for the polyester and vinyl-ester resin used in the GFRP and CFRP samples were approximately 1.1% and 1.6%, respectively. Because these values are smaller than the coefficient of variation (COV), the tensile strength of FRP rebar is not significantly affected by the additional matrix used in the rebar. This minute contribution of the resin type to the improvement of the tensile strength of the FRP rebar is due to the large difference in the tensile strengths of the resin and the fiber. Given that the tensile strength of resin is relatively small compared to that of E-glass fiber (i.e., approximately 77 MPa for resin and 3500 MPa for fiber) [2], the strength of the resin is often ignored when estimating the tensile strength of FRP rebar using Eq. (3), as previously stated.

3.1.3. Fillers

Fillers may be used to control shrinkage and thixotropy in the resin. Also, they are used to improve certain material properties and the load transferring capability of the matrix. Most importantly, they are used to reduce the amount and cost of the matrix used to produce FRP rebar [16]. In order to improve the tensile properties of FRP rebar, the effects of the filler on the GFRP rebar were investigated. Here, three types of filler, consisting of polyvinyl alcohol (PVA), kaolin (KAO), and alumina powder (ALP), were considered. Dog-bone tensile tests were performed on the specimens, as shown in Fig. 2. A FRP specimen made without any filler was the control specimen for comparisons of the tensile strength. A total of

Table 2

Tensile strength test results for FRP rebar with various constituent material combinations.

Composition of bar (fiber/resin)	No. of specimens	Tensile strength (MPa)	Elastic modulus (GPa)
Glass/vinyl ester	1	939	46.5
	2	984	49.8
	3	980	46.2
	4	985	48.8
	Average	972	47.8
	COV	2.3%	3.7%
Glass/polyester	1	1050	47.9
	2	971	48.8
	3	923	48.4
	4	988	48.3
	Average	983	48.4
	COV	5%	1%
Carbon/vinyl ester	1	1389	116.5
	2	1630	118.7
	3	1344	109.3
	4	–	–
	Average	1454	114.8
	COV	10.6%	4.3%
Carbon/polyester	1	1370	113.4
	2	1508	120
	3	1429	112.4
	4	1419	112.6
	Average	1431	114.6
	COV	4.0%	3.2%

40 specimens were prepared (i.e., 10-PVA, 10-KAO, 10-ALP, and 10-control specimens). The average tensile strengths obtained from the dog-bone tests are tabulated in Table 3.

The test results showed that the use of PVA as a filler improved the tensile strength by approximately 6% compared to the control specimens. In contrast, the use of KAO and ALP reduced the strength by approximately 40% and 20%, respectively. Considering that the mix ratio of the filler was relatively small, in the range of 0.4%, the results confirmed that the filler has a strong effect on the tensile strength capacity of FRP. Consequently, a proper type of filler must be selected to optimize the cost and performance.

3.2. Optimization of the manufactured product

3.2.1. Manufacturing process

The common manufacturing processes of FRP are pultrusion, braiding, and filament winding [2]. Among them, pultrusion is the fastest and most cost-effective manufacturing process. This process facilitates the production of axial members such as prestressed tendons and rebars [17]. Also, the process enables the manufacturing of FRP rebar with consistent cross-sections and continuous lengths, constrained only by storage and transporting limitations. Although pultrusion is the most rapid manufacturing process, the rate at which the rebar is pulled through the die is pre-determined by the curing time. Moreover, an additional process is

required to form ribs on the smooth surface of pultruded rebar, which would not provide sufficient concrete bonding without additional ribs. Therefore, additional works are required to improve the bonds between rebar and the surrounding concrete [18]. Ko et al. [19] proposed a new manufacturing process which produces hybrid ductile FRP rebar, as shown in Fig. 3. The new manufacturing process combines the braiding and pultrusion processes and is referred to as the “braiding-pultrusion” or “braidtrusion” process. From this process, continuous ductile FRP rebar can be fabricated at a lower cost compared to the costs incurred when using other composite manufacturing processes. Also, deformed FRP rebar for better bonding in RC structures can be produced simultaneously from an in-line process. However, Ahmadi et al. [20] reported that rebar produced from the braid-pultruded process has a higher shear modulus than that produced from the ordinary pultrusion process. They also reported that FRP rebar with the lowest cover to core fiber weight ratio had a shear modulus approximately 1.5-times higher than rebar produced from the braid-pultruded process. Nevertheless, the tensile elastic modulus and flexural rigidity are lower in braid-pultruded rebar compared to those of unidirectional pultruded rebar when the fiber volume fraction is same. Therefore, modification of the manufacturing process is necessary to improve the tensile strength of FRP rebar produced by the pultrusion process.

To maximize the performance of manufactured FRP rebar, a modified braidtrusion process was used. The process was simplified by eliminating the second resin pool. Also, two types of bobbins or yarn carriers were used during the process, as shown in Fig. 4. One of these was used to induce ribs on bar surface, and the other was intended to cloak the bar surface. This simplified process helps to increase the production rate by completing the production of FRP rebar and the surface configuration in a single continuous processing line, thereby reducing the production cost.

It is important to note that pre-tensioning is applied to the core fibers. Fibers in the core of FRP rebar samples are impregnated by resin through a resin pool. The impregnated fibers pass through a nozzle which has a larger diameter in its entrance than that in its exit. The difference in the diameters between the entrance

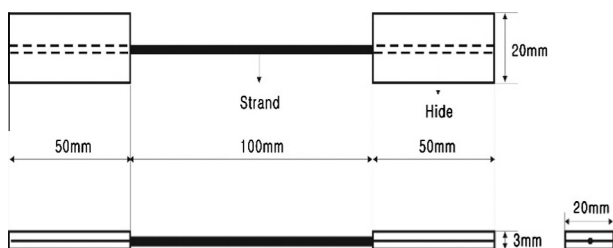


Fig. 2. Details of dog-bone tensile test specimen.

Table 3
Average tensile strength of GFRP dog-bone specimens with various filler materials.

Filler type	Average tensile strength (MPa)	Ratio of performance change
Control	2098	Reference
Polyvinyl alcohol	2220	106%
Kaolin	1316	63%
Alumina powder	1648	79%

and exit generates resistance force against the pulling force created by a puller. Therefore, tensioning of the core fibers is accomplished by these two forces. It is also designed such that the bobbins undergo some friction during the rotation; the strands in the bobbins are pre-tensioned by this friction to tighten the core fibers. The tightening reduces both voids and resin usage in the FRP rebars to enhance the interfacial performance between the fibers. The effect of this tensioning is explained in detail in the next section.

3.2.2. Voids and pre-tension

Earlier results showed that the material properties of FRP rebar can be degraded by air entrainment during the resin impregnation process [21,22]. As stated earlier, the core fibers were tensioned in the modified process, making the fibers straight and pre-stressed. The application of a pre-stress, reduced matrix material is needed to infill the rebar and squeeze out the trapped air in the core. When the pre-stress was released, compressive stress was generated on the resin [23], which produced a confinement effect on the core from the surface-wrapped core fibers. Scanning electronic microscope (SEM) photos of the cross-sections of the FRP rebar are

shown in Fig. 5. The cross-section of a FRP rebar specimen fabricated by a general braiding machine used to manufacture generic products such as rope is shown in Fig. 5(a). The cross-section photo of a FRP rebar specimen produced after modifications were made to the processing apparatus and the addition of tensioning to the core fibers is shown in Fig. 5(b). A comparison of the photos shows the significant reduction in the number of voids from before and after modifications of the braiding process. Finally, after a step to induce ribs on bar surface was added to the process, a further reduction in the number of the voids is achieved, as shown in Fig. 5(b). The reduction in the number of voids improved the interfacial shear bonding between the fibers, and pre-tensioning helped to improve the fiber arrangement. Consequently, these modifications enhanced the tensile properties of the FRP rebar. A SEM photo taken of the cross-section of a GFRP rebar sample produced by the surface strand wrapping of tensioned core fibers is shown in Fig. 6. Because the rebar is made from highly tensioned core fibers with the tight surface wrapping of the fiber strands, the core is tightly packed and large voids are eliminated. Also, the tightly packed core of the GFRP rebar resulted in less filling resin used, thus reducing the cost of the materials.

The tensile strength of GFRP rebar produced after the modification of the apparatus and the implementation of the deformed surface was evaluated according to CSA requirements [24]. A tensile test was performed only on the rebar shown in Figs. 5(b) and 6. The rebar shown in Fig. 5(a) was not tested due to a defect in the cross-section. The tensile strengths of the rebar specimens shown in Figs. 5(b) and 6 were indirectly compared because they were trial products from the GFRP rebar development stage and do not

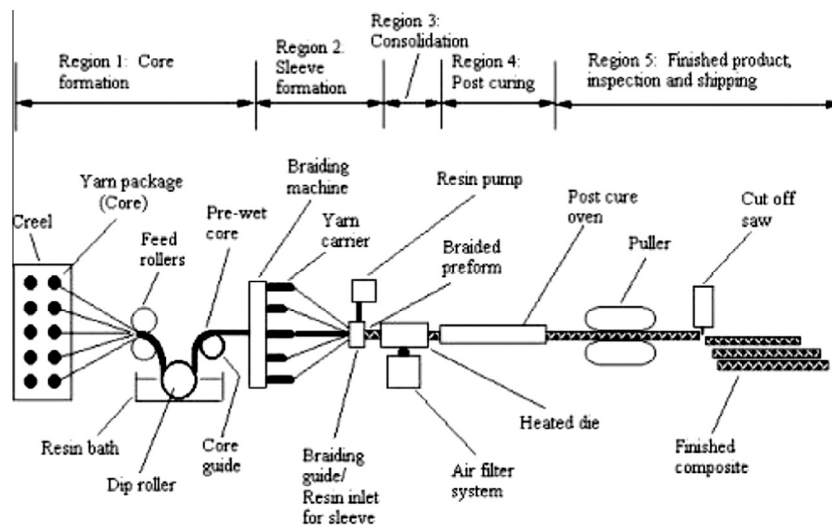


Fig. 3. General braidtrusion process [19].

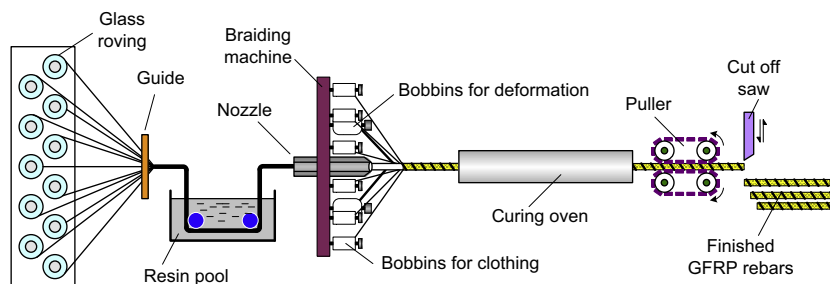


Fig. 4. Modified braidtrusion process.

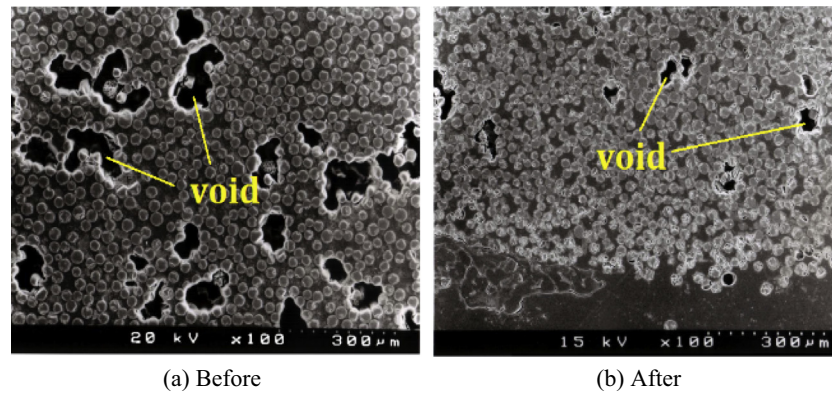


Fig. 5. Microscopic photos of cross-section of GFRP rebar before and after modification of braiding machine.

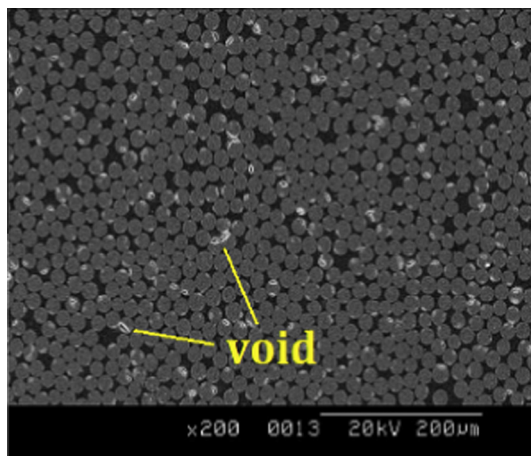


Fig. 6. Microscopic photos of cross-section of GFRP rebar after implementation of surface configuration.

have the same amount of fibers even though the same fiber type was used in rebar manufacturing. Indirect tensile strength is equal to the measured average tensile strength divided by the rove number. The tensile strengths of rebar with the tensioned core fiber and the ribbed surface configuration were 10.03 MPa/rove and 11.92 MPa/rove, respectively. As expected, the reduction in the number of voids improved the tensile strength of GFRP rebar.

3.2.3. Arrangement of core fibers

In order to maximize the performance of FRP rebar, the usage of the constituent materials must be optimized. Although the tensile strength of E-glass fiber is approximately 3500–3600 MPa, the tensile strength of GFRP rebar made using E-glass fiber is approximately 700 MPa [2], which is inadequate considering the volume fraction of the E-glass in the rebar. Various layouts of fibers can occur, as shown in Fig. 7. Twisted, loosened, and unlevelled fibers in FRP rebar contribute to an inadequate load carrying capacity and a reduction in the tensile strength. The tensile strength also decreases if the bonds between the fibers are insufficient, even with proper alignment. A roll of fiber used in the pultrusion process is shown in Fig. 8. Multiple rolls of fiber are roved to produce a FRP rebar specimen. However, due to the non-systematic roving of the fibers, incorrect fiber layouts occur in the final FRP rebar.

The tensile performance of GFRP rebar resulting from the fiber twisting effect was studied. As stated earlier, GFRP rebar is fabricated using multiple fiber rolls (e.g., approximately 100 rolls for a rebar specimen with a diameter of 12.7 mm). Due to the time

and cost constraints of this study, special jigs to ensure the straight roving of the rolls could not be incorporated into the manufacturing process. Therefore, the fiber rolling work for roving was performed manually and 6 mm diameter GFRP rebar specimens excluding rib height were fabricated and tested. The rib height was excluded from the diameter because the rib does not directly contribute to the tensile strength of the rebar. A total of 20 specimens for twisted and untwisted conditions were prepared and tested according to the CSA requirements [24]. Table 4 shows the test results.

The results show that the average tensile strength of the twisted and untwisted GFRP rebar specimens were 1359 MPa and 1363 MPa, respectively. The average tensile strength difference between the untwisted and twisted GFRP rebar samples was exceedingly small, but the increase was approximately 8%. This variation is significant for a rebar with a diameter of 6 mm, because fiber roving is used in all fabrication processes of GFRP rebar. Therefore, the increase in the tensile strength would be much more significant in a GFRP rebar specimen with a diameter greater than 6 mm.

4. Investigation of the performance of optimized GFRP rebar specimens

4.1. Tensile performance

GFRP rebar was fabricated considering the influential factors, as investigated earlier. The rebar in this case consists of E-glass fiber and vinyl-ester resin and has a diameter of 12.7 mm and a volume fraction of 78% by weight. The diameter of the rebar excludes the rib height, because the rib does not significantly contribute to its tensile strength. The volume fraction was measured by burn-out testing. The appearance of the rebar is similar to that of steel rebar, as shown in Fig. 9. To investigate the tensile performance of the GFRP rebar, 15 specimens in total were prepared and tested according to the CSA requirements [24]. The total and gauge lengths of the specimens were 1308 mm and 508 mm (40 times the diameter), respectively. Two steel cylinders of 400 mm length were adapted as a gripping system, which were attached to steel plates by a bolt and nut system, as shown in Fig. 10. The outer diameter and thickness of the steel cylinder were 4.27 mm and 4.9 mm, respectively, which satisfies the requirement of ASTM D7205/7205M 06. An actuator with a capacity of 1000 kN was used for the test. It had a load rate of approximately 450 MPa/min. Strain gauges were installed on the surfaces of the specimens. The GFRP rebar failed in the broom failure mode, as shown in Fig. 11, with perfectly linear behavior, as shown in Fig. 12.

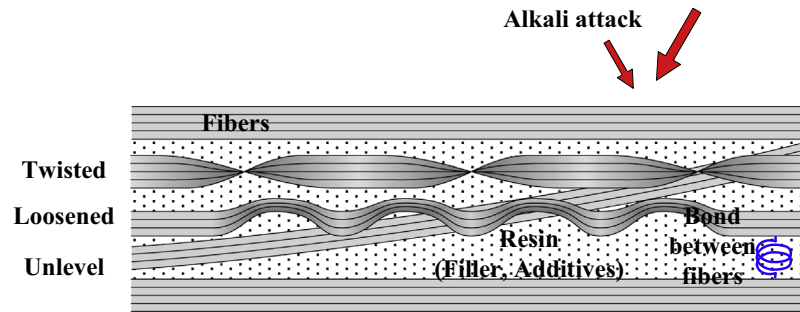


Fig. 7. Various fiber layouts in FRP rebar.



Fig. 8. Roll of glass fiber.

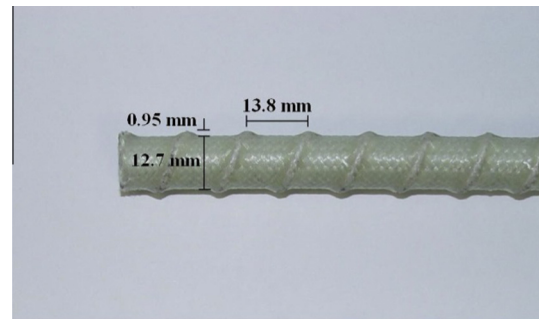


Fig. 9. Deformed GFRP rebar from improved fabrication process.

Table 4
Tensile strength test results for twisted and untwisted GFRP rebars.

Specimen No.	Twisted roving		Untwisted roving	
	Tensile strength (MPa)	Elastic modulus (GPa)	Tensile strength (MPa)	Elastic modulus (GPa)
1	1365	48.6	1375	48.4
2	1396	51.2	1343	50.0
3	1378	49.6	1379	49.4
4	1353	48.7	1306	49.8
5	1370	52.3	1373	50.5
6	1359	49.3	1330	48.7
7	1202	48.6	1347	50.0
8	1355	51.1	1399	50.7
9	1434	49.5	1412	49.5
10	1382	49.9	1367	50.0
Average	1359	49.9	1363	49.7
σ	60		32	
COV	4%		2%	
Average – 3σ	1179		1267	

σ : Standard deviation.

COV: Coefficient of variation.

The tensile strengths and elastic moduli measured from the test are tabulated in Table 5. The average tensile strength and elastic modulus of the GFRP rebar were 1132 MPa and 49.4 GPa, respectively, and the guaranteed tensile strength and ultimate strain were 1059 MPa and 2.02%, respectively. The guaranteed tensile strength of the GFRP rebar showed an increase of approximately 40% over that of conventional GFRP rebar with a tensile strength of approximately 700 MPa. The tensile strength over the volume fraction of the fiber was 17.2 MPa/volume, which is an increase of 46% over commercially available GFRP rebar with a tensile strength of 710 MPa and a fraction of 60.3% by volume and 77% by weight [2,5]. Therefore, it is safe to conclude that the GFRP rebar produced from the optimized manufacturing processes with the constituent materials led to an enhancement of the tensile strength

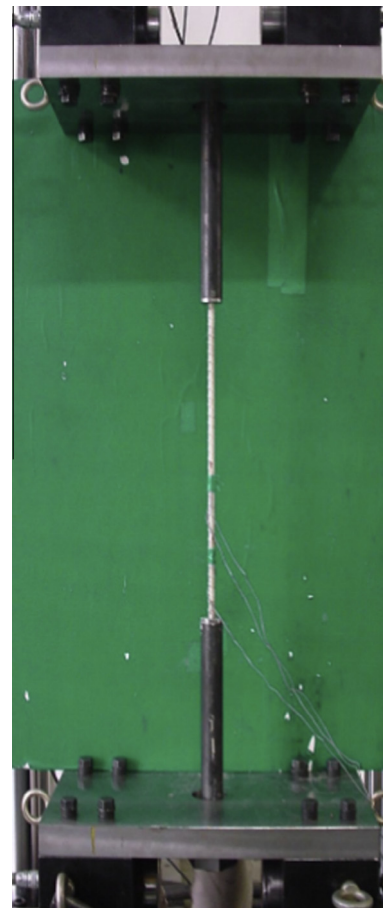


Fig. 10. Tensile test setup.



Fig. 11. Failure mode.

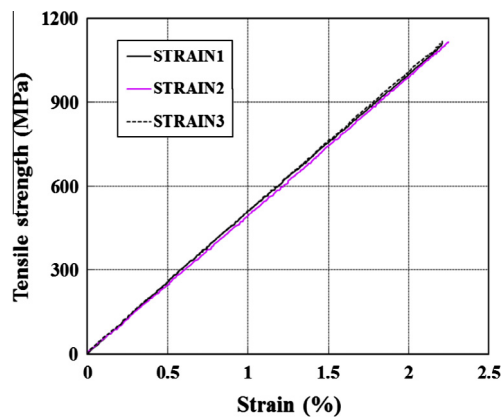


Fig. 12. Tensile stress and strain relationship.

of the rebar. It is important to note that the same constituent material types and amounts used to produce GFRP rebar sold on the market currently were used to produce the new rebar. Ultimately, the target guaranteed tensile strength of 900 MPa was achieved through these optimizations.

4.2. Bond performance

The mandatory requirement for rebar in RC structures is that it must behave uniformly with concrete through good bonding. Therefore, it was necessary to verify the bonding performance of the new GFRP rebar. The bond performance of the GFRP rebar used in this study was evaluated and compared to that of two commer-

Table 5

Tensile strength test results for GFRP rebars.

Specimen No.	Tensile strength (MPa)	Elastic modulus (GPa)	Ultimate strain (%)
1	1147	48.4	2.33
2	1104	50.0	2.20
3	1144	49.7	2.31
4	1182	48.5	2.44
5	1153	49.5	2.31
6	1123	51.3	2.15
7	1112	50.7	2.21
8	1150	48.3	2.32
9	1118	49.5	2.24
10	1125	47.7	2.34
11	1137	49.5	2.24
12	1118	48.8	2.28
13	1166	49.2	2.36
14	1096	48.1	2.22
15	1108	51.1	2.12
Average	1132	49.4	2.27
σ	24.4		0.1
COV	2.2%		3.7%
Average – 3 σ	1059		2.02

 σ : Standard deviation.

COV: Coefficient of variation.

cially available types of GFRP rebar. The GFRP rebar was embedded in a cubic concrete block with a length that was five times the rebar diameter as specified in ACI 440.3R-04 [25]. The diameter of all rebar specimens was 12.7 mm and the compressive strength of the concrete was 33.6 MPa. Specimens were attached to a semi-spherical plate to avoid eccentric loading, as shown in Fig. 13. For comparison, two commercially available GFRP rebar and steel rebar specimens were tested.

The test results are shown in Fig. 14 and Table 6. The x-axis is the relative slip between the displacements of the end of the rebar to the surface of the concrete, and the y-axis is the bond strength calculated by dividing the peak force by the surface area of the embedded length of the rebar. All specimens failed by pull-out, but two of the steel specimens failed by yielding. The steel and V-ROD demonstrated high levels of stiffness in the early stage



Fig. 13. Bond test setup.

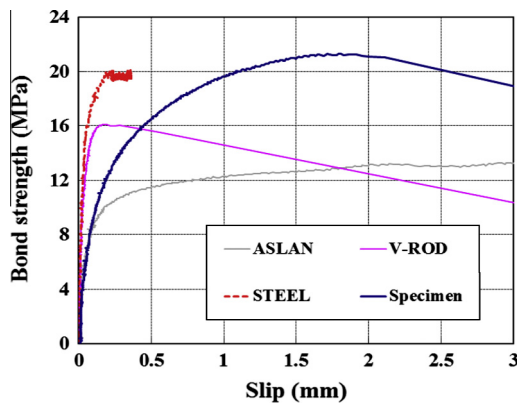


Fig. 14. Bond strength-slip relationship.

Table 6

Bond test results of various GFRP rebars (MPa).

No.	ASLAN	V-ROD	Steel	Specimen in this study
1	12.5	12.0	17.2	21.3
2	14.0	16.1	19.8	18.5
3	15.1	11.5	20.1	18.9
Average	13.8	13.2	19.0	19.6

due to the strong mechanical interlocking noted between the rebar and the concrete. The relative slip of the GFRP specimen was greater than that of the steel specimen, as GFRP is softer than steel. For the V-ROD specimen, the surface was roughened with sand particles, which imparted good mechanical interlocking behavior with the concrete. However, the sand coating ultimately failed in a brittle manner, as shown in Fig. 14. Although the GFRP rebar used in this study had greater relative slip than that of the steel or V-ROD materials, the bond strength was higher than that of commercially available GFRP rebar. These results show that the GFRP rebar enhanced through the aforementioned optimizations can behave uniformly with concrete as one unit.

5. Conclusions

In this study, the manufacturing processes and constituent materials used in the production of GFRP rebar were optimized in an effort to improve the tensile strength while maintaining the production cost for wider application of rebar in the field of construction. The influential factors found from the optimization process were enhanced to produce a new type of GFRP rebar with a minimal guaranteed tensile strength of 900 MPa and good bonding characteristics. The conclusions drawn from the study are as follows:

- (1) The type of resin should be selected properly, but its contribution to the tensile strength of rebar is insignificant.
- (2) The tensile strength of GFRP rebar can be improved by the proper selection of the filler. In addition, a positive effect can be realized when using polyvinyl alcohol filler, as shown in this study.

- (3) Core fiber tensioning and the implementation of a deformed surface reduced the number of voids in GFRP rebar and increased the guaranteed tensile strength to 900 MPa.
- (4) Modification of the roving process to untwist the fibers and maintain a straight alignment greatly improved the tensile strength of the new GFRP rebar.
- (5) The optimized GFRP rebar showed better tensile strength and bond performance levels than the GFRP rebar currently on the market.

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